Manav Rachna International Institute of Research & Studies
School of Engineering and Technology
Department of Computer Science and Engineering
Experiment 5

Course Outcome (CO1): Understand the machine learning algorithms
BTL: L3
Objective: Perform basic data visualization and understand patterns in the dataset using seaborn library

Dataset: Titanic Dataset
Library: Seaborn

Seaborn is a Python library used for data visualization and statistical analysis. It is built on top of Matplotlib and provides a simple and attractive way to create informative statistical graphs.

Seaborn is widely used in Machine Learning and Data Science for Exploratory Data Analysis (EDA), where visualizations help us understand patterns, relationships, distributions, and trends in a dataset.

Key Features of Seaborn

- Provides simple functions for creating statistical visualizations.
- Works easily with Pandas DataFrames.
- Supports numerical and categorical data visualization.
- Provides attractive default themes and color palettes.
- Makes it easy to visualize distributions and relationships between variables.
- Provides tools for correlation analysis, such as heatmaps.
- Helps identify outliers and patterns before applying Machine Learning algorithms.

Common Seaborn Functions

Function	Purpose
countplot()	Display counts of categorical data
barplot()	Compare aggregated values
histplot()	Show data distribution
kdeplot()	Display probability density
scatterplot()	Show relationship between two variables
lineplot()	Show trends
boxplot()	Analyze distribution and outliers
violinplot()	Show distribution and density
heatmap()	Visualize correlation or matrix data
pairplot()	Visualize relationships among multiple variables
jointplot()	Analyze relationship and distributions of two variables
catplot()	Visualize categorical relationships

Practice Questions

1. Load the Titanic dataset and display the first five records.
2. Display the number of passengers belonging to each Passenger Class (Pclass) using an appropriate plot.
3. Visualize the number of male and female passengers.
4. Visualize the number of survived and non-survived passengers.
5. Display the number of passengers in each passenger class based on their gender.
6. Visualize the distribution of Age using a histogram.
7. Visualize the distribution of Fare paid by passengers using a histogram.
8. Display the distribution of Age along with its density curve.
9. Compare the average Fare paid by passengers from different passenger classes.
10. Compare the average Age of passengers belonging to different passenger classes.
11. Visualize the relationship between Age and Fare using a scatter plot.
12. Compare the Age distribution of male and female passengers using a box plot.
13. Visualize the distribution of Fare for different passenger classes using a box plot.
14. Compare the Age distribution of passengers who survived and those who did not survive.
15. Create a violin plot to compare Age distribution across passenger classes.
16. Visualize the relationship between Pclass and Survived using an appropriate categorical plot.
17. Create a heatmap to visualize the correlation among numerical attributes of the Titanic dataset.
18. Create a pair plot using selected numerical attributes such as Age, Fare, Pclass, and Survived.
19. Customize any Seaborn plot by adding an appropriate title, X-axis label, Y-axis label, and legend.
20. Create a visualization of your choice from the Titanic dataset and write 2–3 observations based on the visualization